

PHYSICS PRACTICAL SHEETS

Date: 20.01.2018

PN CAMPUS

Class: BSc 2nd year

Experiment No.: 10

Roll No.: 68

Group: P2

Shift: Day

Sub: Physics

Object of the Experiment (Block Letter) :

Set: 1

TO FIND THE REFRACTIVE INDEX OF THE MATERIAL OF GIVEN PRISM USING A SPECTROMETER.

APPARATUS REQUIRED:

- ① A spectrometer
- ② A spirit level
- ③ A sodium lamp
- ④ A glass prism
- ⑤ An eye-piece

THEORY:

When a ray of light travels from one medium into another, it suffers refraction. If i is the angle of incidence in the first medium and r the corresponding angle of refraction in the second medium, then the refractive index μ of the second medium with respect to the first medium is given by,

$$\mu = \frac{\sin i}{\sin r}$$

According to the wave theory of light,

$$\mu = \frac{\text{velocity of light in vacuum}}{\text{velocity of light in medium}}$$

In the case of prism, when a ray of light passes through a prism it suffers refraction as shown in figure. If EF is the incident ray, FG the refracted ray and GH the emergent ray, then the angle IDH is the angle of deviation. It is the angle between the direction of the incident ray and the emergent ray.

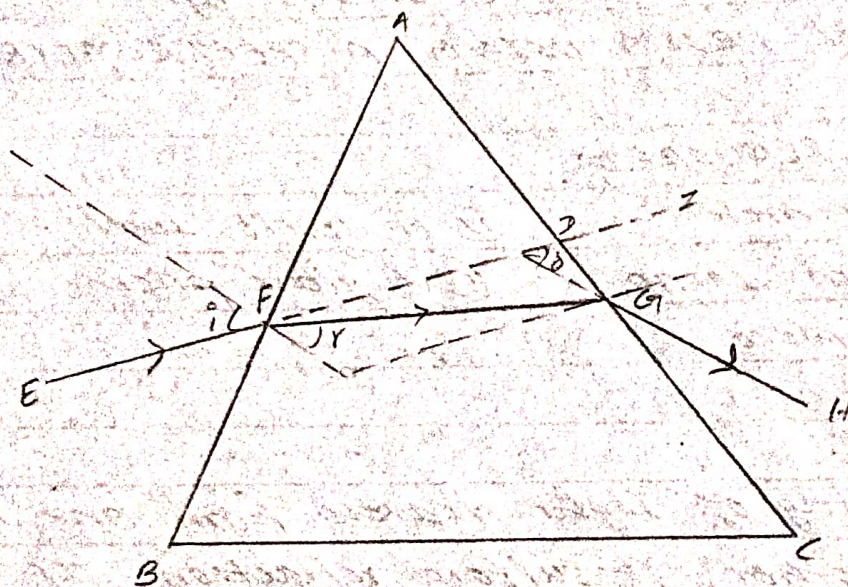


Fig. Refraction in prism

The angle of deviation depends upon the angle of incidence. For a certain value of the angle of incidence, the angle of deviation is minimum. If δ_m denotes the minimum angle of deviation for the given prism of refracting angle A , then the refractive index μ is given by,

$$\mu = \frac{\sin \left(\frac{A + \delta_m}{2} \right)}{\sin \frac{A}{2}}$$

OBSERVATION:

60 vernier scale division (V.S.D) coincides with
59 main scale division (M.S.D).

$$1 \text{ V.S.D} \text{ coincides with } \frac{59}{60} \text{ of M.S.D} = \frac{59}{60} \times \frac{1}{2}$$

Also, 20 small divisions in main scale = 10°

$$1 \text{ small division in main scale} = \frac{10}{20} = \frac{1}{2}^\circ$$

$$\text{Value of } 1 \text{ M.S.D} = \frac{1}{2}^\circ$$

$$\therefore \text{Vernier constant, (V.C)} = (1 \text{ MSD} - 1 \text{ V.S.D})$$

$$= \left(\frac{1}{2} - \frac{59}{60} \times \frac{1}{2} \right)^\circ$$

$$= \left(\frac{1}{120} \right)^\circ$$

$$\text{V.C} = 0.00833^\circ$$

Also,

$$\text{Angle of deviation (A)} = 60^\circ$$

OBSERVATION TABLE:

Telescope reading for V_L .							
S.N	Direct			Min deviation			Difference
	M.S	V.S	Total	M.S	V.S	Total	δm
1.	168	40	168.33	129	0	129.00	$= 168.33 - 129.00$ $\delta m_1 = 39.33$
2.	168	11	168.09	128.5	128	128.59	$= 168.09 - 128.59$ $\delta m_2 = 39.5$
3.	168	19	168.15	129	8	129.06	$= 168.15 - 129.06$ $\delta m_3 = 39.03$

CALCULATION:

From the above table, we have,

$$\delta m_1 = 39.33, \quad \delta m_2 = 39.5, \quad \delta m_3 = 39.03$$

Then,

$$\text{Mean } \delta m = \frac{(39.33 + 39.5 + 39.03)}{3}$$

$$= 117.86 / 3$$

$$\delta m = 39.286^\circ$$

Now,

$$\mu = \frac{\text{sfm} \left(\frac{A + \delta m}{2} \right)}{\text{sfm}(A/2)}$$

$$= \frac{\text{sfm} \left(\frac{60 + 39.286}{2} \right)}{\text{sfm}(60/2)}$$

$$\therefore \mu = 1.524$$

RESULT:

The refractive index of the material of given prism was found to be 1.524.

ERROR PERCENTAGE:

$$\text{error \%} = \left| \frac{\text{standard value} - \text{observed value}}{\text{standard value}} \right| \times 100\%$$

$$= \left| \frac{1.6 - 1.524}{1.6} \right| \times 100\%$$

$$= 4.75\%$$

CONCLUSION:

Hence, the refractive index of the material of given prism is 1.524 with 4.75% error.

PRECAUTIONS:

- (i) The prism should not be touched with wet or greasy hands.
- (ii) Prism should not be moved in middle of observation.